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Architectural Whitepaper & Engineering Specifications

1. Executive Abstract

The deployment of enterprise-grade biometric authentication in field-mobile scenarios necessitates a radical departure from traditional, cloud-dependent topologies. The Offline Secure Biometric Authentication Framework (OSBAF) is a proprietary, edge-native architecture custom-built to operate asynchronously alongside the NHAIDatalake 3.0 ecosystem. Designed strictly for disconnected, high-latency environments, OSBAF shifts the entire cryptographic and inferential workload directly to the mobile edge. This whitepaper details the structural engineering of OSBAF, outlining its highly constrained neural footprint, heuristic anti-spoofing layer, cryptographic storage methodologies, and self-optimizing telemetry synchronization engine.

2. Core Model & Embedded Footprint Optimization

Deploying deep convolutional neural networks (CNNs) on commercial-off-the-shelf (COTS) mobile hardware introduces strict constraints on memory utilization, thermal throttling, and storage boundaries. To maintain institutional agility and strictly adhere to the **<20MB hackathon limit**, the OSBAF architecture abandons monolithic model deployments in favor of a highly quantized inference engine.

The core pipeline is driven by the **InsightFace 'buffalo_sc' ArcFace model**. Through aggressive parameter pruning and dynamic quantization, the compiled model binary maintains a lean storage footprint of **exactly 16.3 MB**. This sub-20MB profile ensures rapid over-the-air (OTA) deployment and minimal heap allocation during runtime.

To mitigate variable computational overhead, the framework enforces a **static grid constraint of 320x320 pixels** for all ingress visual tensors. By preventing dynamic spatial reallocation and restricting the tensor dimensions, the model ensures deterministic memory bounding. This strict 320x320 normalization inherently minimizes raw CPU cycle overhead, mathematically guaranteeing a fully resolved identity inference in **under 0.8 seconds without active GPU acceleration or dedicated NPU availability**.

3. Geometric Liveness Criteria: The "Zero-Byte Liveness Layer"

Presentation Attack Detection (PAD) traditionally requires a secondary, heavily weighted CNN, which fundamentally violates strict mobile footprint constraints. OSBAF engineers around this limitation via the introduction of the "**Zero-Byte Liveness Layer**"—a deterministic, heuristic algorithm that requires zero additional neural network weights.

Instead of pixel-level artifact detection, OSBAF utilizes the **YuNet face detector** to extract precise **2D facial keypoint landmarks (kps)**. The anti-spoofing mechanism relies on topological geometry rather than semantic segmentation. The mathematical engine tracks the horizontal translation of the *nasal apex* relative to the stationary *inter-pupillary distance* (the axis between the left and right ocular centers).

By calculating the dynamic ratio of these delta values, the algorithm accurately derives **2D skull yaw metrics** in real-time. To definitively prove liveness against flat-plane attacks (e.g., holding up a high-resolution photograph or a tablet screen), the engine issues a **randomized challenge validation sequence** requiring the subject to sequentially articulate their cranial pitch and yaw to specific programmatic vectors: **LEFT, RIGHT, and CENTER**. The geometric shift in the nasal-pupillary ratio confirms active, three-dimensional spatial manipulation, bypassing the need for computationally expensive depth sensors or secondary PAD models.

4. Local Cryptographic Storage Vault

Given the sensitive nature of biometric data operating within the NHAIDatalake 3.0 ecosystem, edge persistence is treated as an actively hostile environment. The OSBAF persistence layer utilizes an embedded local SQLite database initialized as **face_auth.db**, operating under a strict zero-trust data lifecycle.

Upon successful visual capture, the **raw photo data is immediately dropped from the volatile heap post-inference**. No pixel data is ever committed to the physical disk, achieving privacy-by-design through absolute data destruction. What remains is a mathematical abstraction: the **512-dimensional vector facial embeddings** generated by the ArcFace model. To persist these vectors securely, the float arrays are serialized natively using Python's **pickle** protocol. The resulting byte stream is immediately intercepted by a cryptographic pipeline and encrypted using **256-bit AES via the symmetric Fernet protocol**. Only the encrypted cipher-text is subsequently committed to the SQLite database. It is stored specifically as an **encrypted Binary Large Object (BLOB)**, ensuring that even if the mobile device is compromised via physical root access, the biometric templates remain cryptographically opaque and mathematically irreversible.

5. Asynchronous Cloud Synchronization & Disk Purge Engine

To bridge the gap between offline operational capability and the centralized NHAIDatalake 3.0 infrastructure, OSBAF implements a highly autonomous, fault-tolerant background synchronization module.

The edge node utilizes persistent hardware and browser state listeners to detect active internet connectivity. Upon a verified positive network state, the background engine queries the **face_auth.db** for un-synced telemetry and audit logs explicitly marked with the **'PENDING'** state flag.

Rather than executing expensive, single-record HTTP requests, the engine aggregates these

pending records into a highly compressed, **unified JSON batch payload**. This payload is transmitted asynchronously to secure **AWS endpoints over TLS 1.3 HTTPS**. The lifecycle of edge data relies on explicit state consensus. Crucially, the local records remain locked until the AWS cloud endpoint returns a definitive HTTP 200 execution acknowledgment. Upon receipt of this ACK, the edge engine aggressively **purges local records** from the SQLite tables. To prevent physical flash memory fragmentation and mitigate write-amplification on mobile NAND storage, the process terminates by triggering a standard **SQLite 'VACUUM' routine**, physically rebuilding the database file, reclaiming zeroed sectors, and optimizing the application's ongoing memory footprint.

6. Technical Performance Profiles

The following tables summarize the operational specifications of the OSBAF architecture under baseline field conditions.

6.1. Neural & Computational Profile

Parameter	Specification	Constraint Metric
Primary Inference Model	InsightFace 'buffalo_sc' (ArcFace)	Industry Standard
Storage Footprint	16.3 MB	< 20 MB Hackathon Limit
Input Tensor Dimensionality	320x320 Pixels (Static)	Deterministic RAM Bound
Feature Extraction Vector	512-Dimensional Float	High-Fidelity Cosine Similarity
CPU Inference Latency	< 0.8 Seconds	Sub-Second UX Requirement
Liveness Extractor	YuNet 2D Keypoint (kps) Detection	Zero-Byte Overhead
Geometric Verification	Nasal Apex vs. Inter-pupillary Ratio	3D Spatial Confirmation

6.2. Cryptographic & Telemetry Profile

Parameter	Specification	Security Mechanism
Local Datastore	SQLite (face_auth.db)	ACID Compliant
Raw Media Retention	0 Bytes	Immediate Volatile Purge
Serialization Protocol	Pickle (Byte-stream)	Data Structuring
Encryption Standard	256-bit AES (Symmetric Fernet)	Military-Grade Encryption
Database Datatype	Encrypted BLOB	SQL Parsing Resistance
Telemetry Transport	Unified JSON Batch (HTTPS to AWS)	TLS 1.3
Garbage Collection	SQLite VACUUM post-ACK	Flash Memory Optimization

Conclusion

The OSBAF framework successfully demonstrates that high-security, cryptographically sound biometric authentication can be achieved strictly at the edge without heavy infrastructural reliance. By merging the 16MB InsightFace architecture with a mathematically derived Zero-Byte Liveness layer and a self-optimizing AES-256 storage vault, OSBAF delivers an uncompromised, zero-trust gateway tailored for the discontinuous realities of the NHAIDatalake 3.0 ecosystem.